

Scenario 1: Delivery Time Analysis for an E-commerce Company

An e-commerce company tracks delivery times (in minutes) for 15 orders:

[25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95]

The company wants to analyze the delivery performance using percentiles and detect if there are any unusual delivery times.

Question 1:

Calculate Q1 and Q3.

Answer:

```
Q1 = np.percentile(data, 25)
```

```
Q3 = np.percentile(data, 75)
```

Question 2:

Find the Interquartile Range (IQR).

Answer:

```
IQR=Q3-Q1
```

Question 3:

Detect Outliers using the IQR method.

Answer:

```
IQR=Q3-Q1
```

```
Lower outlier range = Q1-1.5*IQR
```

```
Higher outlier range = Q3+1.5*IQR
```

```
Outliers = Data < Lower outlier range or Data > Higher outlier range
```

Scenario 2: Student Score Analysis

A teacher is analyzing the mathematics scores of students in her class. The scores are:

[45, 50, 55, 60, 60, 62, 63, 65, 90, 95]

Question 1:

Calculate the mean, median, and mode of the scores.

Answer:

```
from statistics import mean, median, mode
```

```
data= [45, 50, 55, 60, 60, 62, 63, 65, 90, 95]
```

```
mean= mean(data)
mode=mode(data)
median=median(data)
print(f"Mean:{mean},Median:{median} and Mode:{mode}")
```

Mean:64.5,Median:61.0 and Mode:60

Question 2:

Explain why the median might be a better representation than the mean in this case.

Answer:

Median might be the best option as it will omit the outliers(90,95)

Scenario 3: Grocery Store Customer Analysis

A grocery store manager tracks how many customers visit the store daily for a month:

[5, 10, 8, 15, 20, 5, 12, 14, 10, 18]

Question 1:

Create a frequency distribution table for this data.

Answer:

```
import pandas as pd
data= [5, 10, 8, 15, 20, 5, 12, 14, 10, 18]
df=pd.DataFrame(data)
df.value_counts().sort_index()
```

5	2
8	1
10	2
12	1
14	1
15	1
18	1
20	1

Scenario 4: Real Estate Model Analysis

A real estate model has three variables:

- House Size
- Number of Rooms
- Number of Bathrooms

Question 1:

How can you detect multicollinearity in this model?

Answer:

Multi collinearity exists in this dataset. Because the variables - Number of Rooms and Number of Bathrooms can have almost same data (like 4 rooms and 4 bathrooms)

Scenario 5: Medicine Effectiveness Study

A company made a new medicine to lower blood pressure. They gave it to one group and gave a fake pill (placebo) to another group.

Question 1:

How can the company check if the new medicine works?

Answer:

To check the two different group of patients, Independent - Unpaired t-Test can be used to detect the medicine works or not.

Step1: Using `ttest_ind`, find the `p_value`.

Step2: If $p \leq 0.05$, H_0 rejected which means Medicine works.

Else if $p > 0.05$, H_0 accepted which means Medicine fails.

Scenario 6: Identifying Outliers in Sales Data

A company wants to find any unusual spikes in sales.

Question 1:

How can the company detect outliers in their sales data?

Answer:

Method1 :

Using 1.5 rule(Inter Quartile Range), find lower and higher range of outliers.

$IQR = Q3 - Q1$

Lower outlier range = $Q1 - 1.5 * IQR$

Higher outlier range = $Q3 + 1.5 * IQR$

Outliers = Data < Lower outlier range or Data > Higher outlier range

Method 2: Using Box plot, where the datapoints beyond the whiskers are flagged to be outliers

Scenario 7: Understanding Customer Satisfaction

A restaurant conducted a survey to rate customer satisfaction on a scale of 1 to 5:

[5, 4, 4, 5, 3, 4, 5, 2, 4, 3]

Question 1:

How can the restaurant summarize the overall satisfaction?

Answer:

```
from statistics import mean
data=[5, 4, 4, 5, 3, 4, 5, 2, 4, 3]
mean= mean(data)
mean =3.9
```

This indicates the restaurant has high level of customer satisfaction